The on-detector is comprised by 896 Minidrawers (MDs), in which data from all TileCal PMTs will continuously be sampled at 40 MHz. Each MD (figure 1) will host up to twelve channels by means of twelve PMTs that turn light pulses to electric signals to be shaped and conditioned by Front-End Boards. A Mainboard (MB) will continuously sample and digitize two gains of shaped signals of the MD PMTs. A Daughterboard (DB) is in charge of distributing LHC synchronized timing, configuration and control to the front-end, and continuously transmitting the digital data from all the MB channels to the off-detector systems via multi-Gbps optical links.

2 ATLAS TileCal link Daughterboard

The DB is a radiation tolerant board with a redundancy layer of two functionally-equivalent halves, interconnecting the on- and off-detector by means of a 400-pin FMC connector and multi-Gbps optical links, respectively [3]. The DB, currently on its sixth revision (DB6, figure 2), powers the optical links by means of four SFP+ modules that drive two downlinks and four uplinks.

Taking into consideration the position of the DB in ATLAS, commercial off-the-shelf components have been used for the design following the availability of the required components in the market, with the exception of the CERN radiation hard GBTx ASICs [4], due to the importance of the stable clock recovery and distribution very specific to the LHC readout systems. Reported radiation tests studies performed in the ProASIC [6] and the Kintex Ultrascale FPGAs [7, 8] in addition to the tests performed internally in the collaboration were taken into consideration for the selection of the aforementioned components due to their complex nature. The remaining part of the COTS were tested up the radiation requirements in parallel to the board development process.

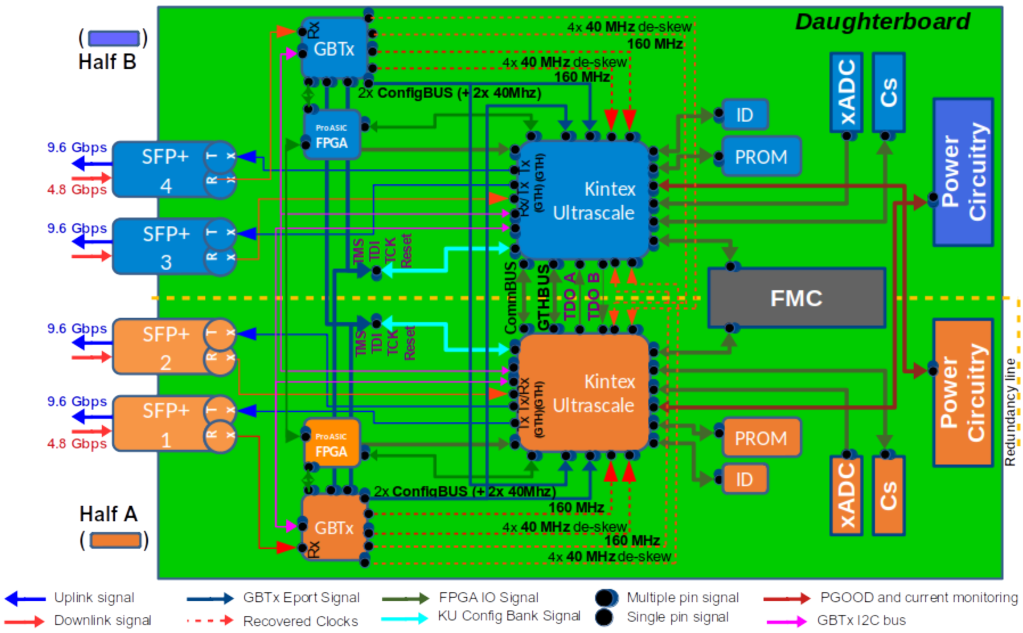

Figure 2. The Daughterboard revision 6 block diagram.

The information received by the board from the off-detector by means of 4.8 Gbps GBT-FEC downlinks powered by CERN radiation hard GBTx ASICs includes clock signals, time

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